Negative acceleration does not always mean slowing

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Average acceleration / ES: aceleración media / PT: aceleração média

Change in velocity divided by elapsed time.

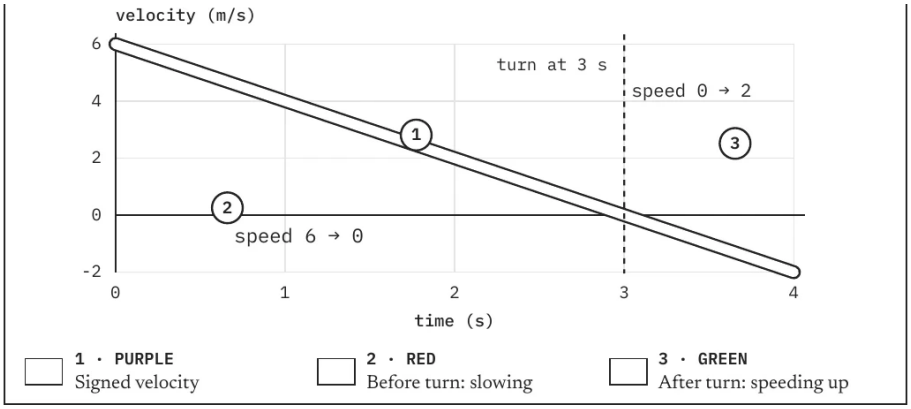
Also on your note page: Speeding up, Slowing down, Reversal.

B READ THE RELATIONSHIP

Average acceleration = change in velocity ÷ time

As you read, find: before · after · change.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A cart's velocity changes uniformly from +6 m/s to −2 m/s in 4 s . Find its acceleration and the time it reverses.

Given $v_i = +6$ m/s $v_f = -2$ m/s $\Delta t = 4$ s

Rule $a = \Delta v / \Delta t$. Turn: $0 = v_i + at$.

Work $a = (-2 - 6)/4$ $0 = 6 - 2t$ $t = 3$ s

Answer $a = -2$ m/s² Slows for 3 s , then speeds up to the left.

E SAVE FOR THE FINAL MEETING

A change in XJ's velocity requires an acceleration term in the forecast. Its sign alone does not show whether the asteroid is speeding up; compare the directions of velocity and acceleration.

Read the physics: Average acceleration

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Compare velocity before and after. A change in speed or a turn is a change in velocity. Divide the signed change in velocity by the time it took.

02 HOW IT WORKS

Read m/s² as a change in meters per second each second. Look at dot spacing at equal times: growing gaps mean faster motion; shrinking gaps mean slower motion.

03 WHAT TO CHECK

Compare velocity and acceleration. In a straight line, matching directions mean speed grows. Opposite directions mean speed falls until the object stops or reverses.

04 TRY THE EXAMPLE FROM YOUR NOTES

A cart's velocity changes uniformly from +6 m/s to −2 m/s in 4 s . Find its acceleration and the time it reverses. a = −2 m/s² Slows for 3 s , then speeds up to the left.

Model check: Average acceleration = change in velocity ÷ time.

05 CONNECT TO XJ

A change in XJ's velocity requires an acceleration term in the forecast. Its sign alone does not show whether the asteroid is speeding up; compare the directions of velocity and acceleration.

SAY IT BEFORE YOU WRITE

Can acceleration be negative while speed increases? The velocity changes from ___ to ___. Each second it changes by ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Average acceleration

ES: aceleración media / PT: aceleração média

Meaning in my words	How can you tell?
How would you explain average acceleration to a friend? In my words, this means ___.	Which two velocity readings would you subtract, and in what order? I can tell because ___.
Example or sketch	**Non-example and why**
A cart changes from −2 to +4 m/s in 3 s. What is its average acceleration? My example shows ___.	A partner divides the whole distance by time. What quantity did that give? I would change ___ because ___.

Speeding up

ES: aumento de rapidez / PT: aumento de rapidez

Meaning in my words	How can you tell?
How would you explain speeding up to a friend? In my words, this means ___.	A cart changes from −2 to −5 m/s. What happens to its speed? I can tell because ___.
Example or sketch	**Non-example and why**
Draw three equal-time dots for a cart speeding up to the right. My example shows ___.	A cart moves right with acceleration to the left. Is it speeding up before it turns? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Slowing down

ES: disminución de rapidez / PT: redução de rapidez

Meaning in my words	How can you tell?
How would you explain slowing down to a friend? In my words, this means ___.	Velocity changes from −5 to −2 m/s. What happens to the speed? I can tell because ___.
Example or sketch	**Non-example and why**
Draw velocity and acceleration arrows for a bike braking while moving right. My example shows ___.	"Negative acceleration always means slowing down." What happens if velocity is negative too? I would change ___ because ___.

Reversal

ES: inversión del sentido / PT: inversão do sentido

Meaning in my words	How can you tell?
How would you explain reversal to a friend? In my words, this means ___.	On a velocity–time graph, how can you tell a cart actually reversed? I can tell because ___.
Example or sketch	**Non-example and why**
A cart goes from +2 m/s through zero to −1 m/s. Draw its direction before and after. My example shows ___.	A bus stops, then keeps going east. Does stopping prove it reversed? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw the before and after velocity arrows (m/s). Mark the change in velocity (m/s). Divide by time to label the acceleration (m/s²).

Labels you can use: before · after · change.

Before

After

Tell a partner

The velocity changes from ___ to ___. Each second it changes by ___.

Choose the speed alert

07

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use a velocity change to find acceleration.

CHOOSE ONE JOB TOGETHER

TELEMETRY ENGINEER ☐

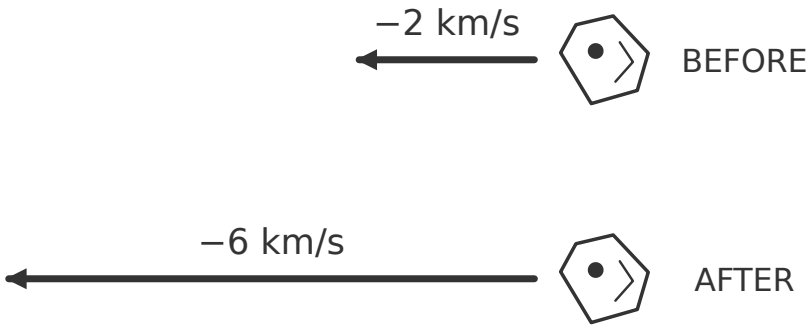
Radio the average acceleration with its sign and unit. Label both velocity arrows so the crew can check it.

REPLAY COMMENTATOR ☐

Narrate the two-second change. Point to the arrow lengths to explain whether XJ sped up or slowed down.

Word help: velocity / acceleration / speeding up

Two signed XJ velocities are **−2** and **−6 km/s**, two seconds apart. The control room must choose its speed alert.



01 OUR FIRST CALL

Which speed alert should the control room use? Support it with the arrows.

The speed ___ because the arrow ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the proposed banner. Give a corrected report using the two velocities.

Now we would say ___ because ___.

Switch speakers. Why do two velocity readings give an average acceleration rather than the acceleration at every instant?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks